CIS678 - Machine Learning

– Syllabus –

Instructor: Yong Zhuang

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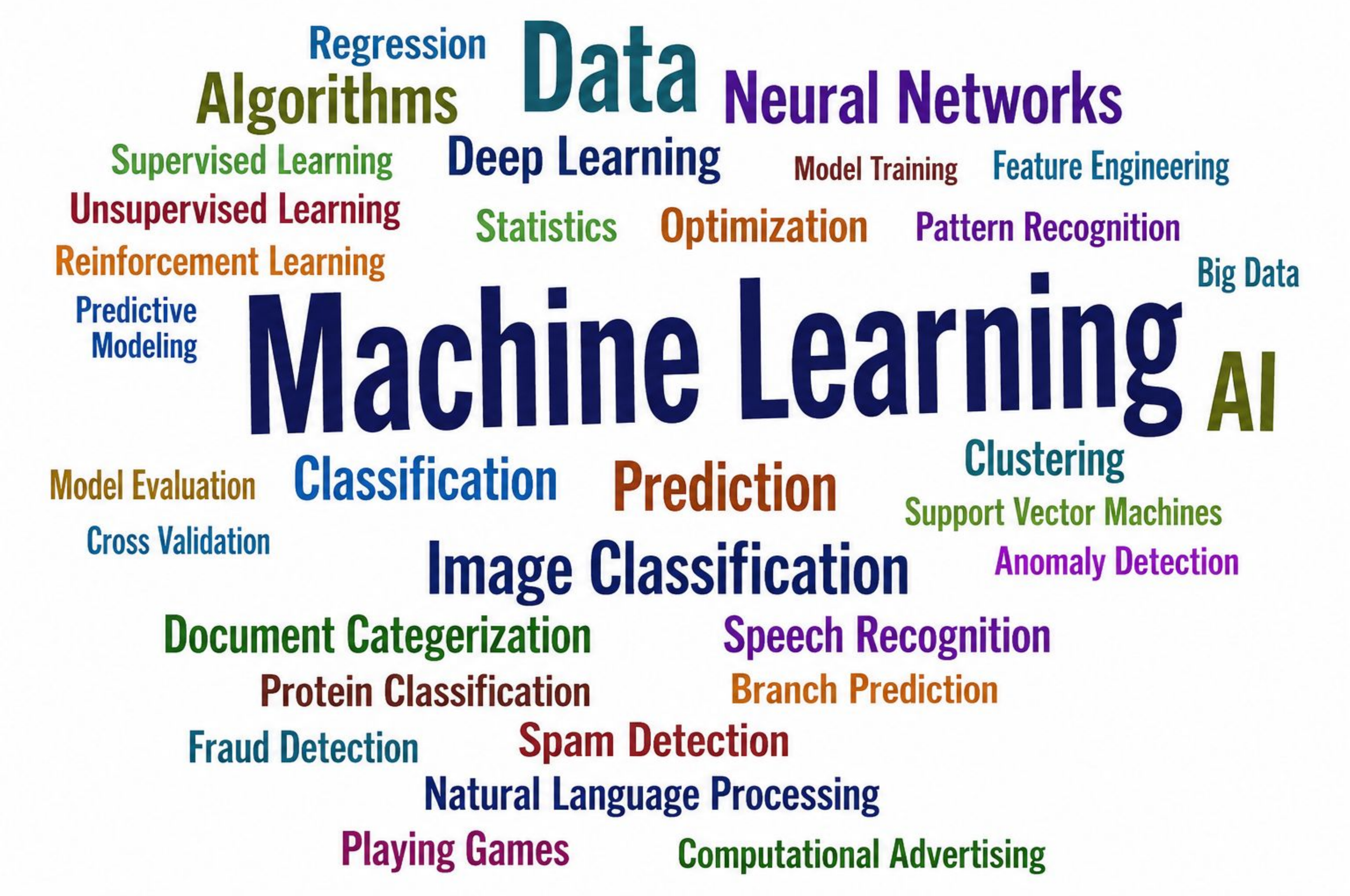
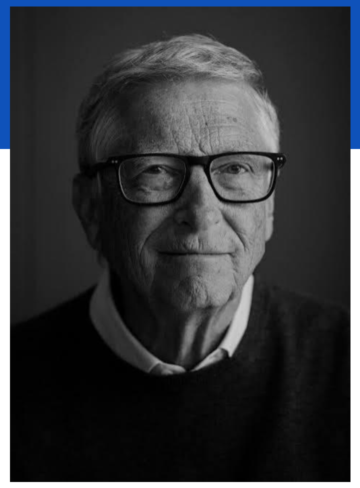
Machine Learning

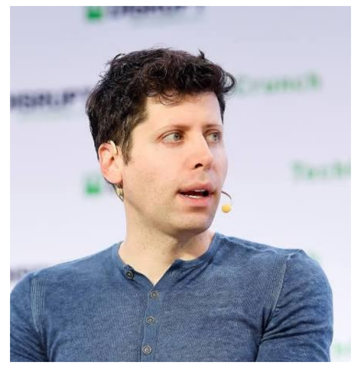


The COOLEST TOPIC IN SCIENCE

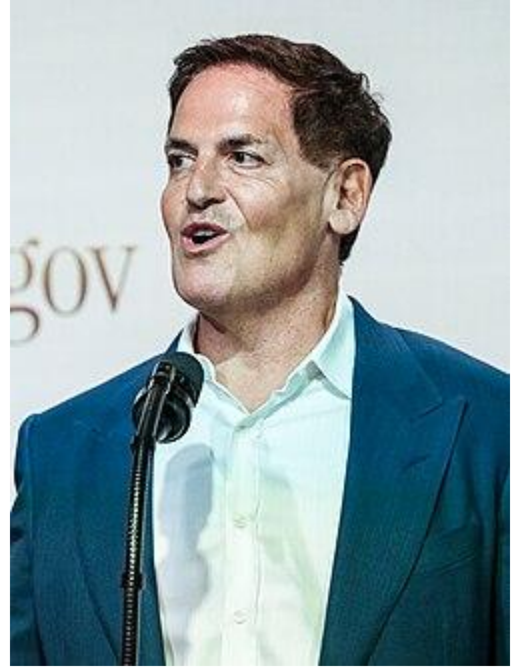


- **Bill Gates:** “A breakthrough in machine learning would be worth ten Microsofts.”

- **John McCarthy:** “Our ultimate objective is to make programs that learn from their experience as effectively as humans do.”



- **Sam Altman:** “AI won't replace humans, but humans who use AI will replace those who don't.”



- **Mark Cuban:** “Artificial Intelligence, deep learning, machine learning — whatever you're doing if you don't understand it — learn it. Because otherwise you're going to be a dinosaur within 3 years.”
- **Alan Turing:** “I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.”

About this course

Cover (some of) the most commonly used machine learning paradigms and algorithms.

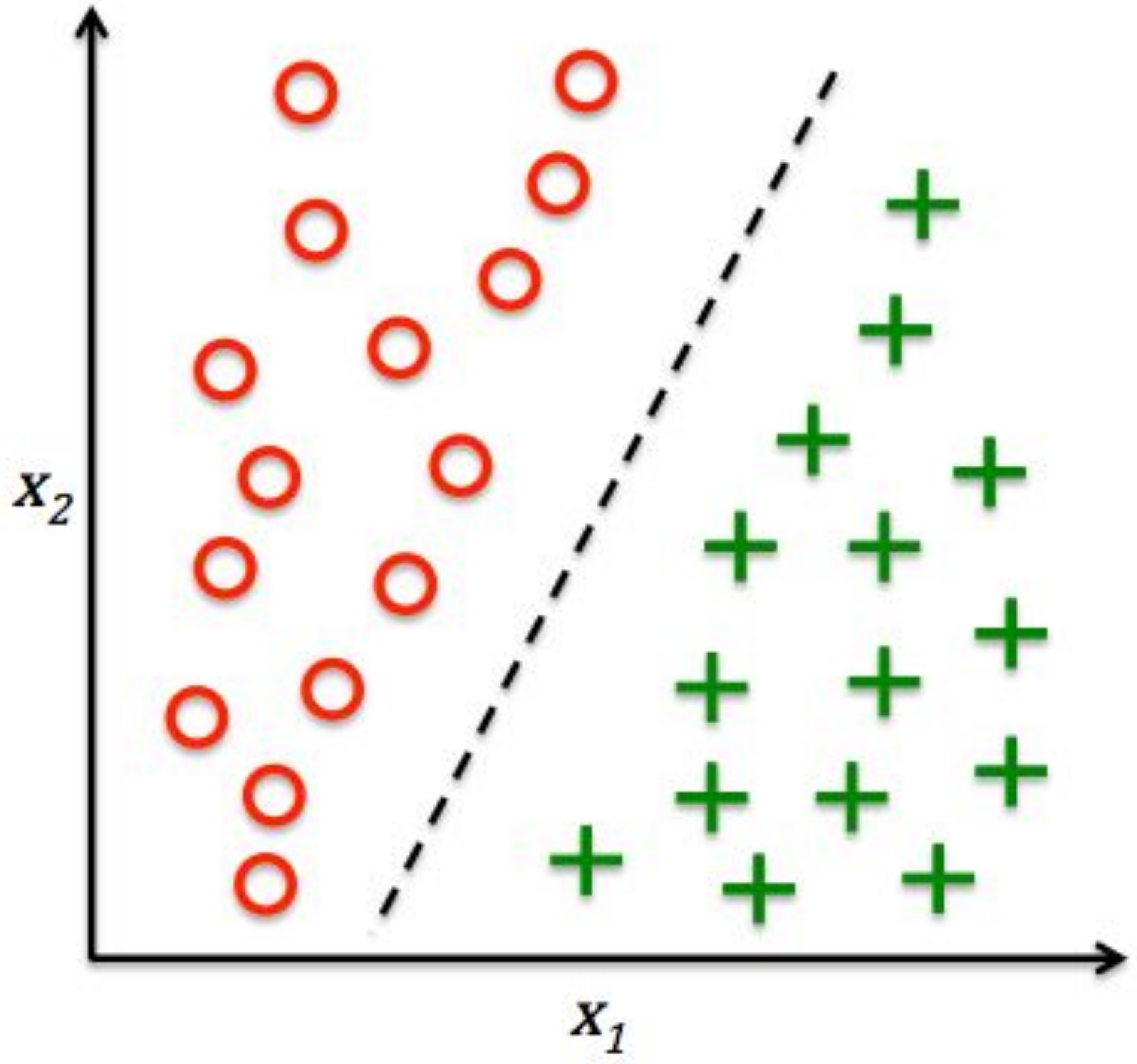
- Sufficient amount of details on their mechanisms: explain why they work, not only how to use them.
- Applications

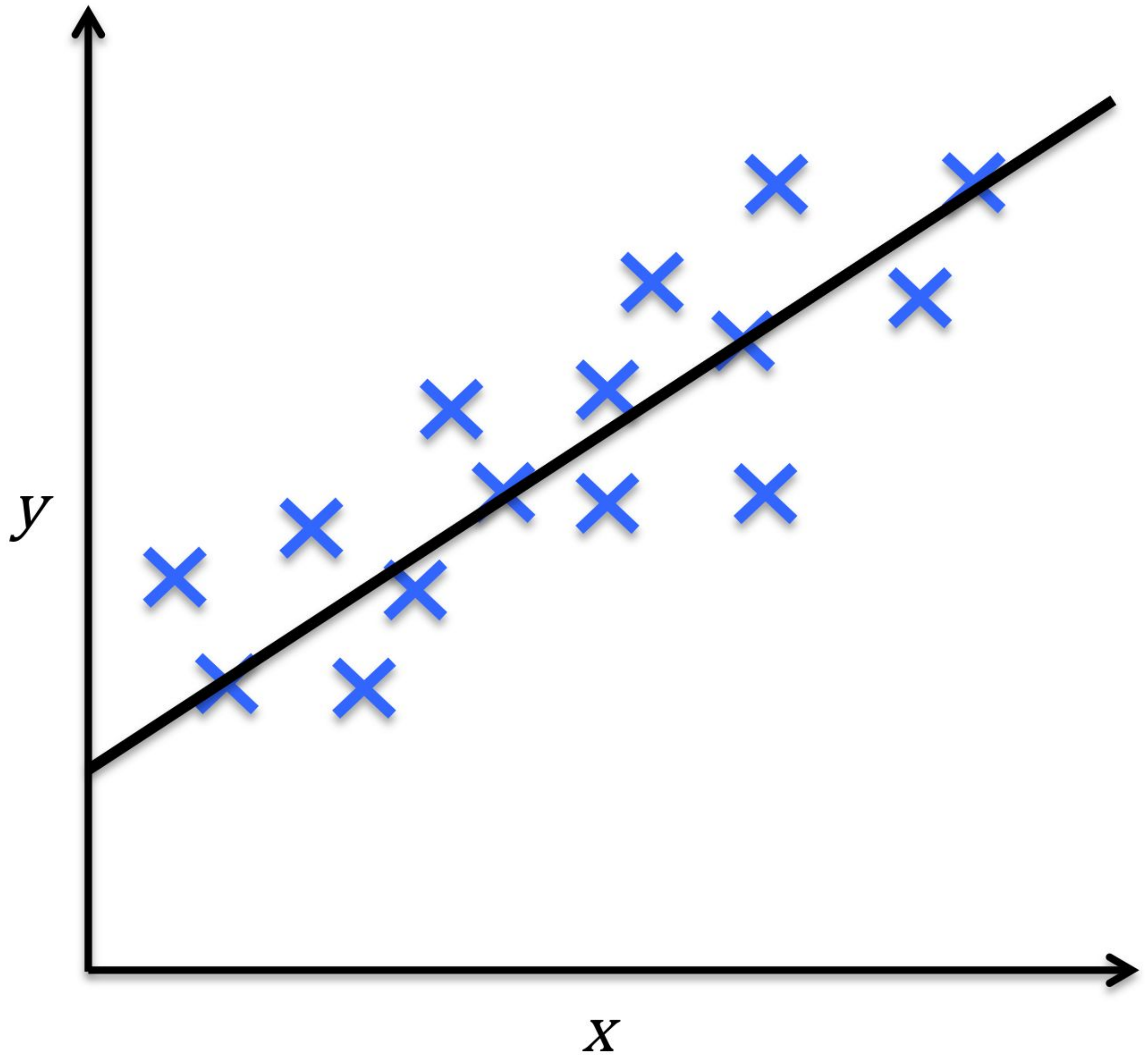
What is Machine Learning

Examples of important machine learning paradigms.

What is Machine Learning

- Supervised Learning 
  - Data: both the features, x, and a target, y, for each item in the dataset
  - Goal: ‘learn’ how to predict the target from the features, y = f(x)
  - Example: Regression and Classification





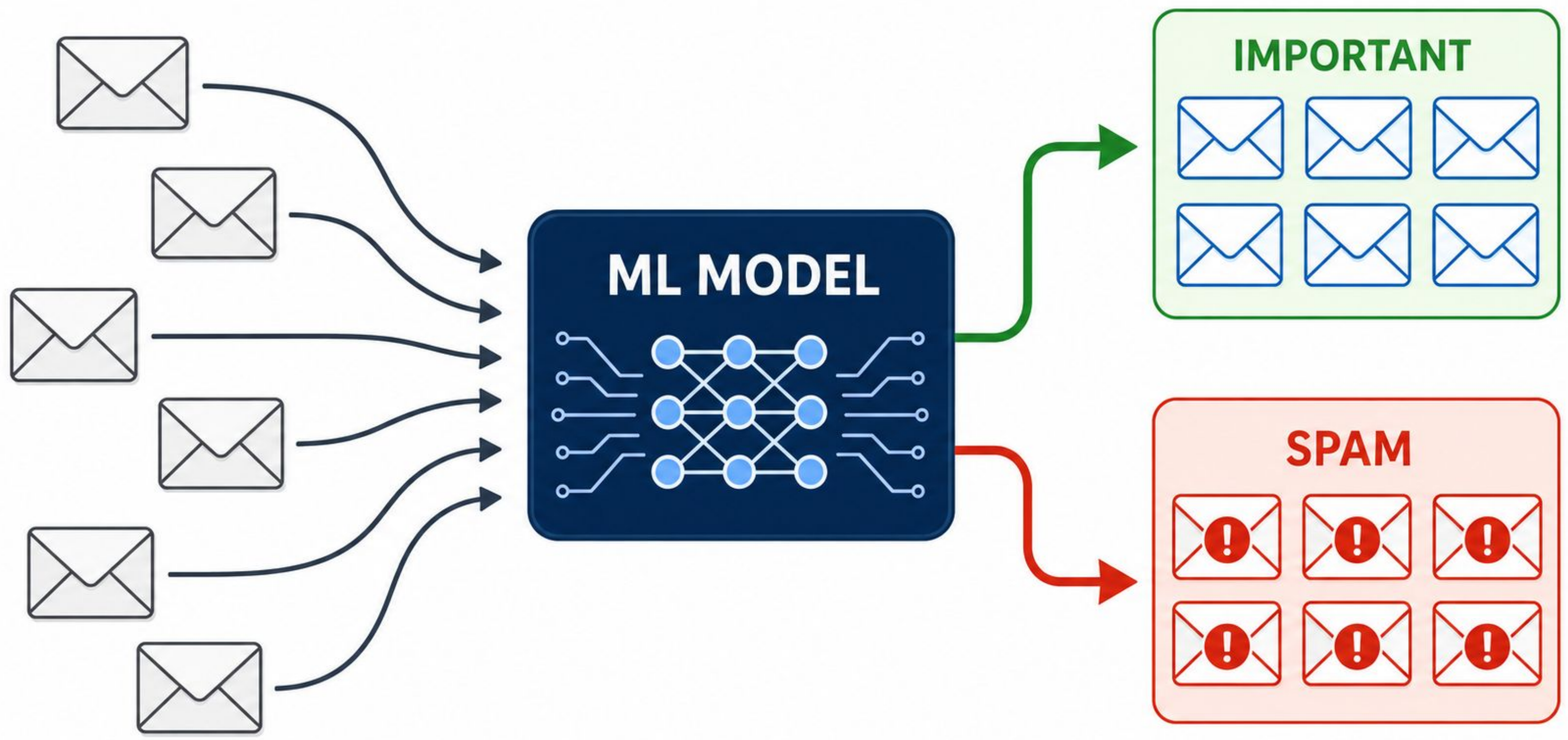
What is Machine Learning

Supervised Classification

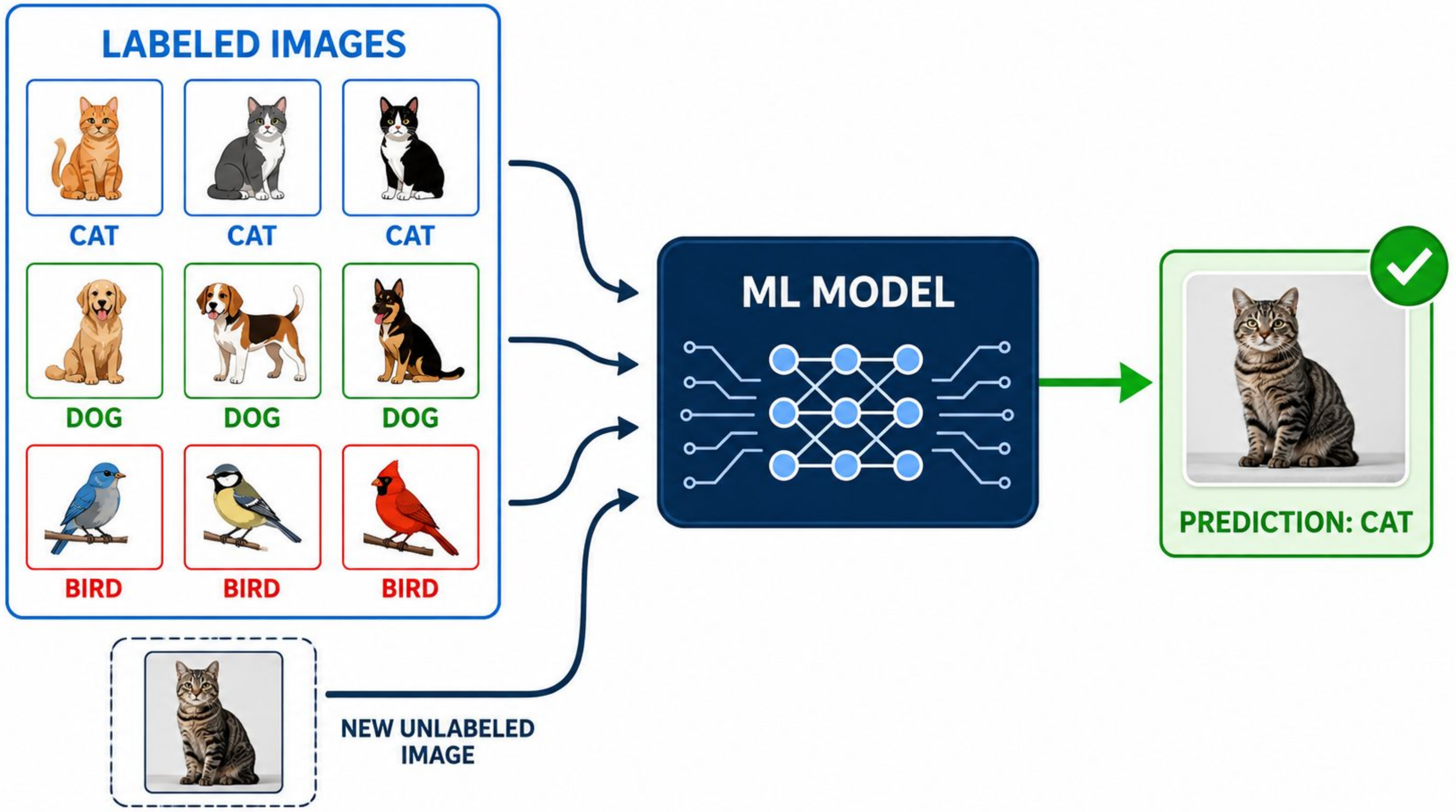
from data to discrete classes

Supervised Classification. Example: Spam Detection

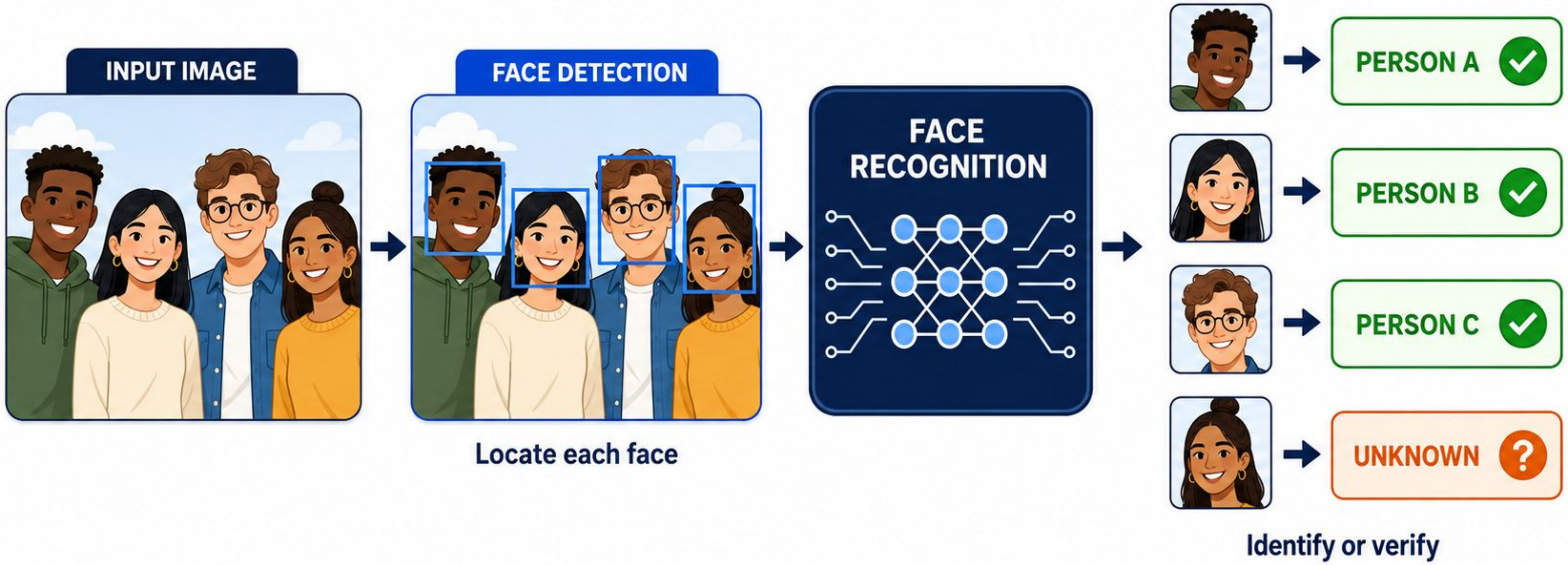
Decide which emails are spam and which are important.



Supervised Classification. Example: Image classification

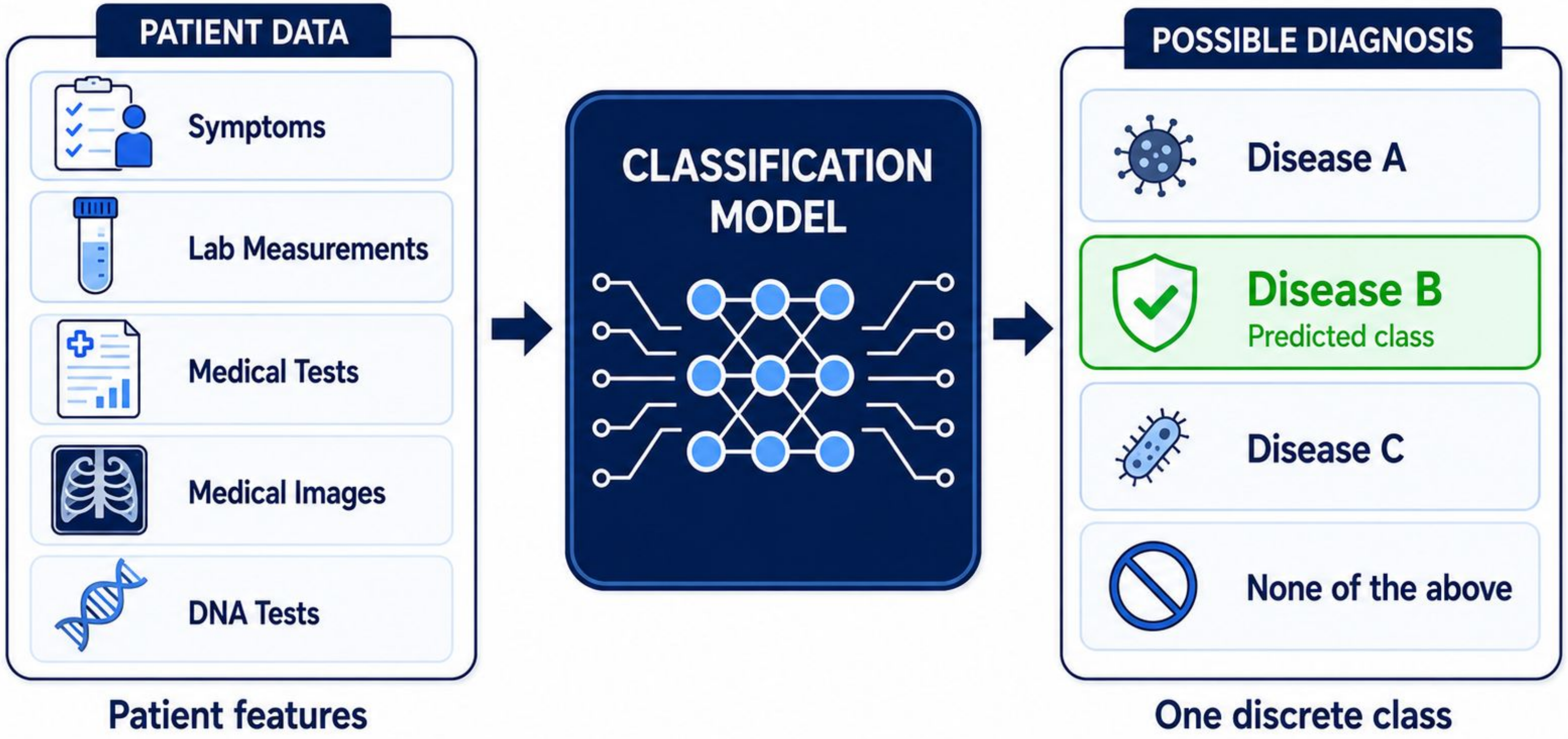


Supervised Classification. Example: Face Detection, Recognition



Face detection: locating faces with bounding boxes
Face recognition: identifying each detected person or marking them as unknown

Supervised Classification. Example: Disease Diagnosis

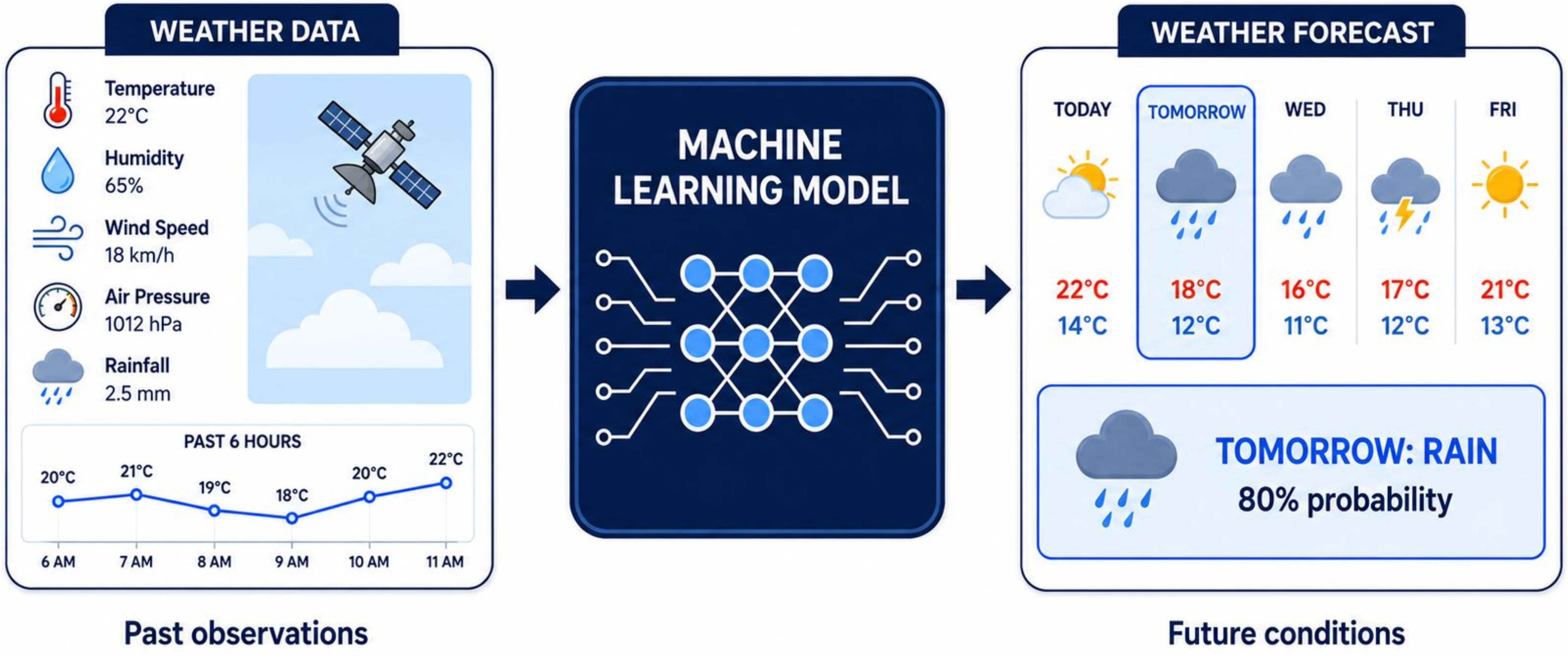


What is Machine Learning

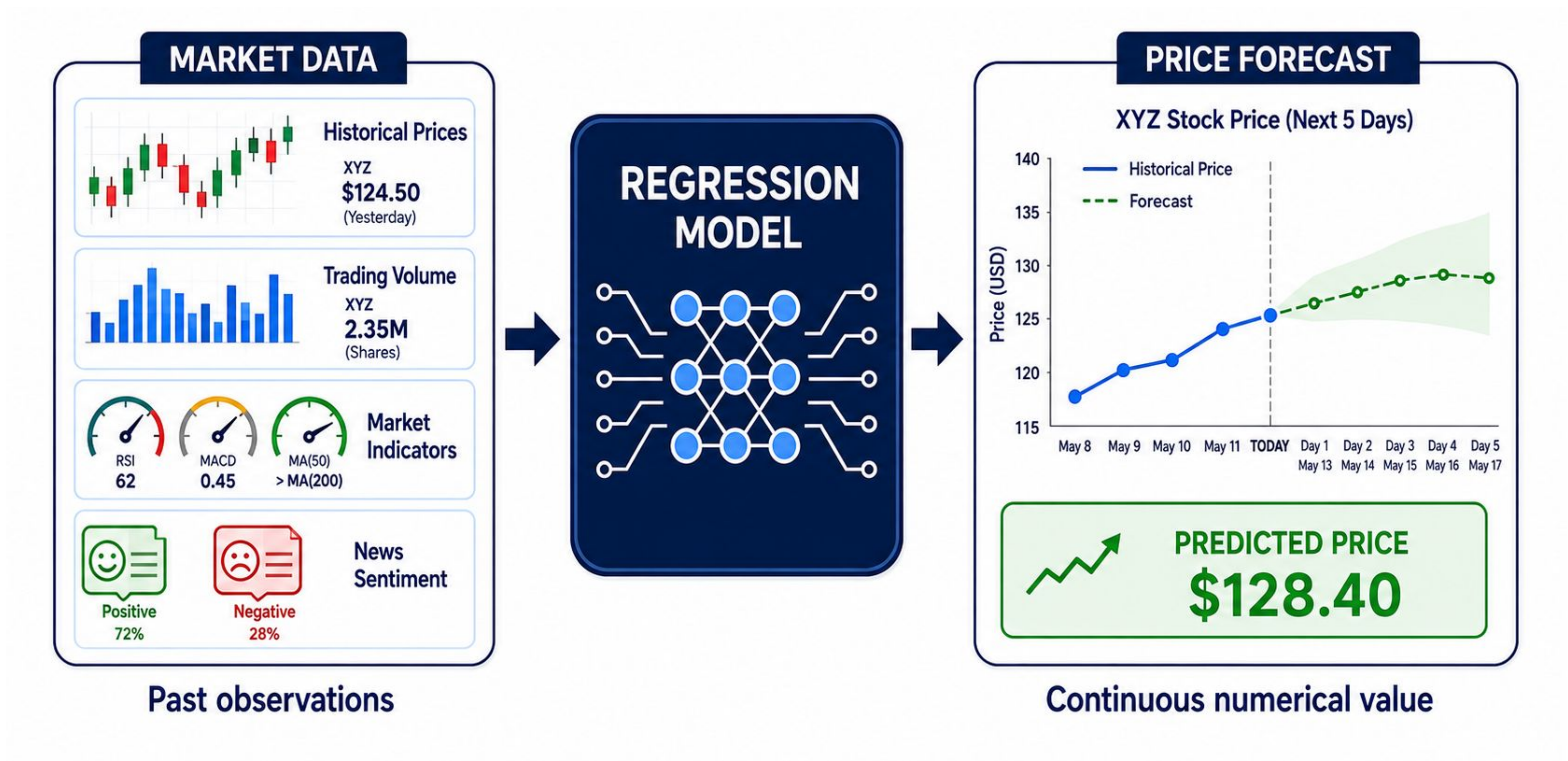
Supervised Regression

from data to continuous numerical values

Supervised Regression. Example: Weather Prediction

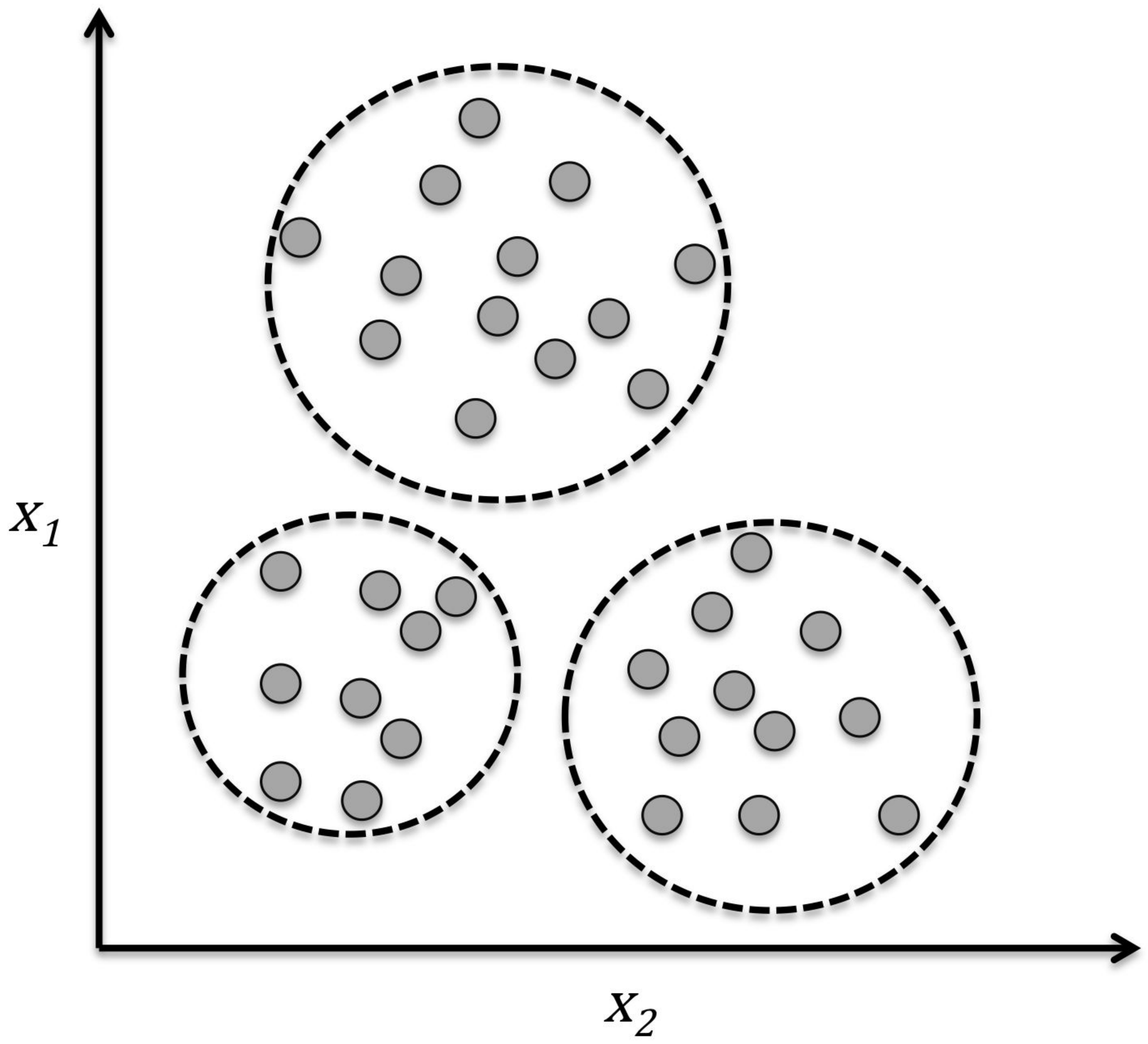


Supervised Regression. Example: Computational Economics

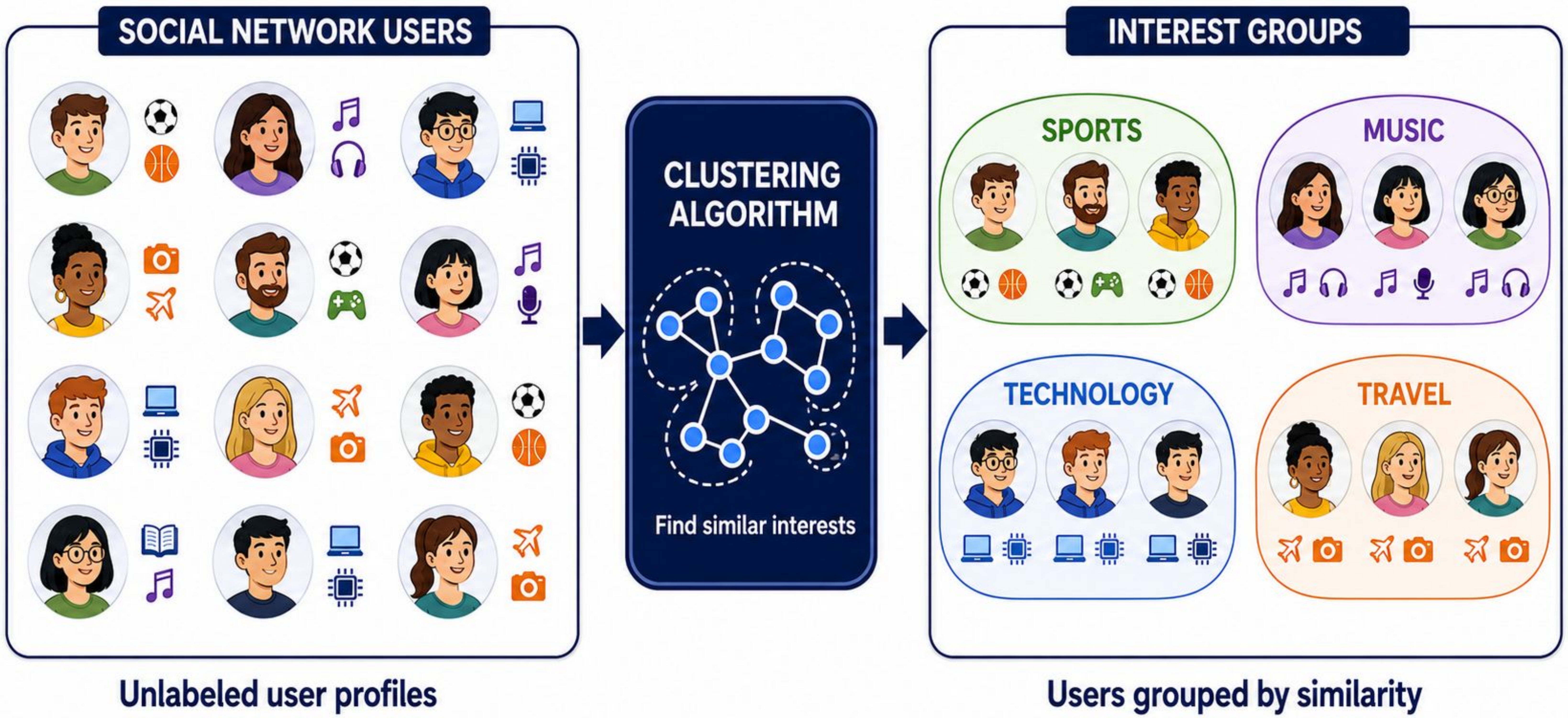


What is Machine Learning

- Unsupervised Learning 
  - Data: Only the features, x, for each item in the dataset
  - Goal: discover 'interesting' things about the dataset
  - Example: Clustering, Dimensionality reduction,



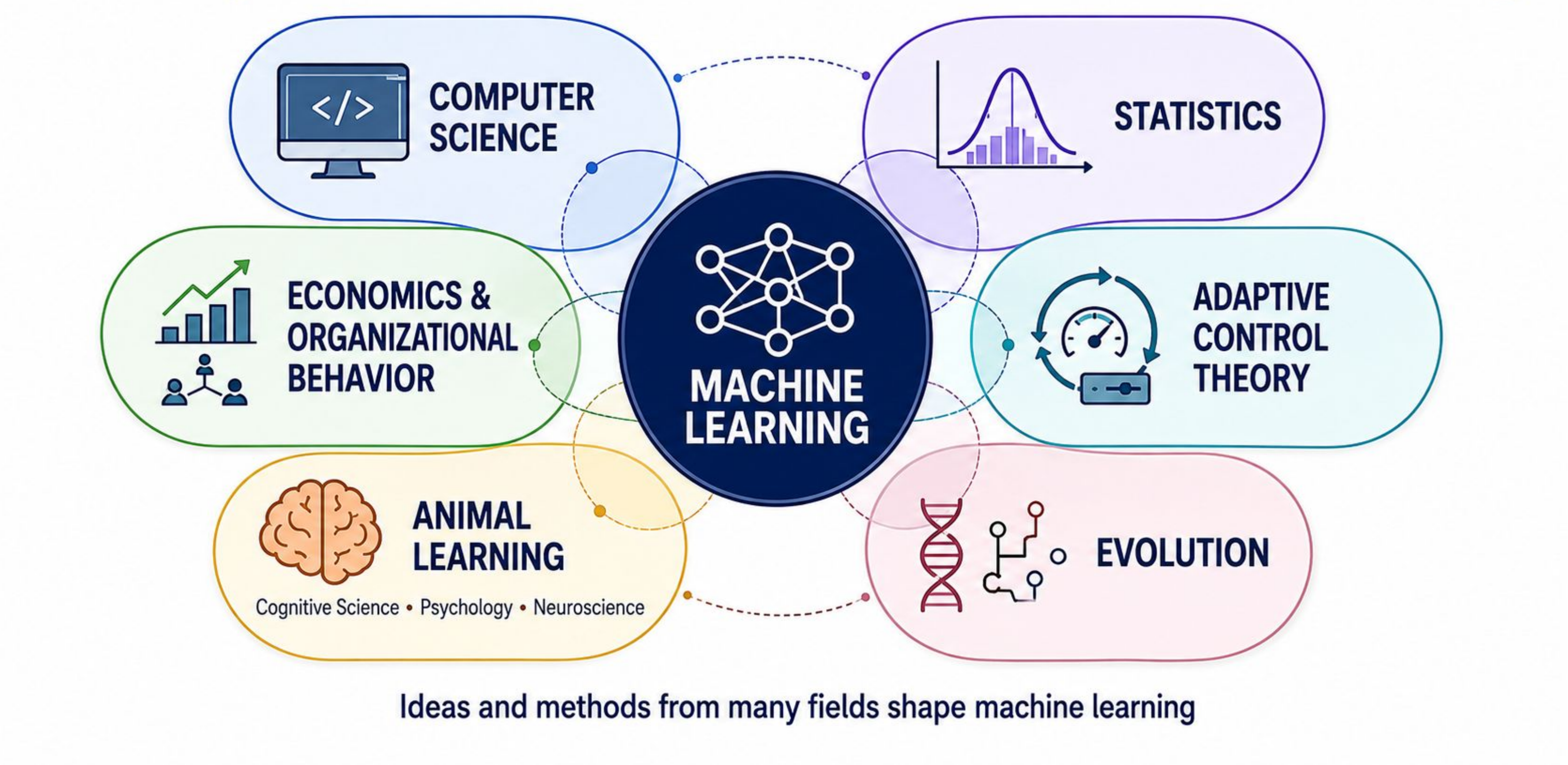
Unsupervised Learning. Example: Clustering



Other Machine Learning Paradigms

- Semi-Supervised Learning: Learns from a small amount of labeled data and a larger amount of unlabeled data.
- Self-Supervised Learning: Creates training targets from unlabeled data and learns useful representations.
- Active Learning**:** Selects the most informative unlabeled examples and requests labels from a human expert.
- Reinforcement Learning**:** Learns actions through rewards or penalties, including indirect or delayed feedback.
- Collaborative Filtering: predicts user preferences from patterns in user–item interactions.
- …

Many communities relate to Machine Learning



About this Course

Admin, Logistics, Grading

Contact Information

- Instructor: Dr. Yong Zhuang
- You can call me Dr. Zhuang (draw on), or Yong (you own)
- Homepage: https://yong-zhuang.github.io
- E-mail: yong.zhuang@gvsu.edu
- Office: MAK D-2-234
- Office hour: Monday 1:00 pm - 3:00 pm, MAK D-2-234
  - Or by appointment. Send me an email to schedule a time to meet (over Zoom or in-person).
  - https://gvsu-edu.zoom.us/j/3966686420?pwd=WGxpc0N4YWcvOU9aWGxWZGYxbXZUdz09
  - Meeting ID: 396 668 6420
  - Passcode: 587684

Course Information

- Course Homepage:
  - Blackboard: https://lms.gvsu.edu/
  - Course Website: https://gvsu-cis678.github.io
- Prerequisites
  - CIS 500 or CIS 661 or equivalent.
  - Basic programming experience and introductory knowledge of probability, statistics, Algorithms, Big O notation, and linear algebra(matrices and vectors).
    - *We may review these things but we will not teach them*

Course Objectives

After completing this course, students should be able to:

- Explain fundamental machine learning concepts and algorithms and describe their use in prediction and data-driven knowledge discovery.
- Implement and modify established machine learning algorithms and design extensions or new learning approaches.
- Compare machine learning algorithms, explain their assumptions, strengths, and limitations, and relate selected approaches to human learning processes.
- Develop end-to-end machine learning solutions involving data preparation, feature transformation, model selection, hyperparameter tuning, evaluation, and error analysis.
- Apply machine learning methods to real-world datasets and communicate experimental results through appropriate quantitative and visual analysis.
- Explain current developments and research directions in machine learning.

Textbook

There are no required textbooks for this course. If you would like additional references, some good options include:

- Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn and PyTorch*, O'Reilly Media, 2025.
- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, *An Introduction to Statistical Learning with Applications in Python*, 2023.
- Tom Mitchell, *Machine Learning*, McGraw-Hill, 1997.
- Christopher Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, MIT Press, 2012.
- Slides, recommended readings, online references, and other resources will be made available on the course website and Blackboard.

Grading

- Quizzes & Homework assignments 30%
- Project 30%
- Midterm 20%
- Final Exam 20%

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Grading: Homework

Homework: 30%

- Homework assignments are designed to reinforce course material and include a mix of programming tasks along with mathematical or written questions.
- Deadline: 11:59 pm Michigan time on the due date.
  - *Late policy*: Assignments submitted late will incur a 10% penalty per day, capped at five days (50%). After this period, the assignment **will not be accepted**.
- **No copying or sharing of homework!**
  - But you can discuss general challenges and ideas with others

Grading: Course Project

Course project: 30%

- Group project (3-4 people for one group)
- Goal: Solve a given machine learning problem, prepare the data, compare at least two machine learning approaches, use an appropriate validation strategy, analyze the results, and discuss model limitations.
- You are expected to submit a project report and your code at the end of the semester

Academic Honesty

- Document all collaborations.
- No electronic code transfers between students.
- Code you find on the internet must be cited, with an active link to that code. That code should not solve the entirety of an assigned problem/project (i.e., don't have someone else do your project for you).
- You are encouraged to engage in conversations in online forums, but do not post solutions or solicit others to complete your work for you.
- You are encouraged to talk about problems with each other in non-technical terms (i.e., not code)
- Generative AI Tools: Unless an assignment states otherwise, generative AI tools may be used for explanation, brainstorming, and debugging. Any substantive use must be disclosed and cited. Students may not submit AI-generated code, analysis, or writing that they do not understand and cannot explain. Generative AI tools may not be used during quizzes or exams unless explicitly permitted. Students remain responsible for the correctness, originality, security, and academic integrity of all submitted work.
- Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.

- **Suspicious cases will be reported to the Academic Honesty Committee**

Tentative Course Content

- **September 6 - 7, Labor Day Recess:** No classes!
- **October 26-27, Fall Break:** No classes!
- **November 25-29, Thanksgiving Recess:** No classes!

Week	Topics Covered
1	Machine learning foundations, learning paradigms, and software tools
2	Labor Day Recess (No Class)
3	Data preparation, model evaluation, and applied machine learning workflow
4	Statistical learning and regression methods
5	Classification methods
6	Decision trees and ensemble learning
7	Similarity-based and kernel learning methods
8	Midterm exam
9	Fall Break (No Class)
10	Bayesian learning methods
11	Unsupervised learning and dimensionality reduction
12	Neural networks and learning frameworks
13	Genetic algorithms and explanation-based learning
14	Reinforcement learning methods
15	Current machine learning research, Project presentations
16	Final exam

Expectations for Success

- Check Blackboard on a regular basis for announcements and assignments
- Check https://gvsu-cis678.github.io for course material.
- I'm here to help you! Come to office hours or schedule an appointment when you're feeling lost/stuck.
- Let me know when something is getting in the way of your success in this class.
- Let me know how the class and my teaching can be improved.
- Adhere to class, CIS, and GVSU policies on academic honesty.

Resources

GVSU provides opportunities for students to improve your academic skills through resources, such as

- Writing center (https://www.gvsu.edu/wc/)
- Speech lab (https://www.gvsu.edu/speechlab/)
- Research consultants (https://www.gvsu.edu/csce/)
- Library liaisons (https://www.gvsu.edu/library/about-the-university-libraries-3.html)
- Academic success center (https://www.gvsu.edu/sasc)

Warming Up

Individual introduction

- Name and what do you want to be called
- Describe your background and experience in Machine Learning or AI.
- Specific topics you seek to learn from this class

About Me

- You can call me Dr. Zhuang (draw on), or Yong (you own)
- Education:
  - Ph.D., M.S. in Computer Science from the University of Massachusetts
- Experience/Interests:
  - My research interests include web application design, machine learning, and data mining.
- Personal Interests:
  - I enjoy spending time in city parks, hiking, and playing board games with friends and family.